Note: This Rubrics is for the course CLOUD COMPUTING (CSIZG527/SEZG527/SSZG527)

Cloud Computing Evaluation Criteria

Read more about Evaluation Rubrics for phases 1, 2, and 3 in the Evaluation Templates

Core Information:

Instructor: Dr. SHWETHA VITTAL
Course: Cloud Computing.
Number of Tasks: 3
LF: Dr. Sumanth Reddy

Learning Objectives (By the end of these tasks, Students will be able to):

1. Design and justify a cloud-based solution aligned with Situation Learning (SL) guidelines, ensuring workplace relevance, open-ended problem framing, data sensitivity handling, feasibility, and reflective learning
2. Proper documentation of the work.
3. Present the key contributions and learnings toward the executed work.
4. Coordinate with team members.

Project Type:

- ❏ Group Project (Up to Members)

Time Estimate:

PHASES	DESCRIPTION	DURATION	DELIVERABLES
Phase 1	Problem Statement Definition and Justification	17-03-2026 to 31-03-2026	Initial Presentation + PPT Submission
Phase 2	Implementation and Submission of Deliverable s	03-04-2026 to 03-05-2026	A. PPT, B. Recording of Working Project, C.Code/Scripting done D. Report Submission

Phase 3	Final Presentation	04-05-2026 to 10-05-2026	Presentation + Q&A Session (Schedules will be provided)

- Students must not send assignment documents or presentation files through email under any circumstances.
- All presentation slides (PPT/PDF) must be submitted only through the Taxila LMS portal within the specified deadline.
- Email submissions will not be accepted for evaluation.
- It is the responsibility of the student to ensure successful upload on Taxila and verify submission status before the deadline.

Overview

Since students can have access to various cloud platforms, this project allows implementation on AWS, Azure, GCP, or any other cloud service. However, as this course demonstrates the use of AWS exclusively, students must provide a service mapping if they use a different platform (e.g., Azure, GCP).

Assignment Overview:

Situation Learning (SL) Guidelines for Problem Statement

The problem statement must align with the following SL criteria:

1. Relevance to Workplace

The problem should mirror real challenges faced in an organization such as infrastructure upgrades, user onboarding, performance issues, scalability limitations, or seasonal traffic spikes.

2. Authentic and Contextualized Learning

The problem must be grounded in a real business case. Students should design solutions that address practical challenges related to scalability, availability, and performance in cloud environments.

3. Open-Ended Problem Statement

The problem should be broad and exploratory without a single fixed solution. Students must evaluate multiple cloud service options, compare alternatives, and justify architectural decisions.

4. Handling Data Sensitivity

The solution must address secure management of sensitive data, including compliance, encryption, access control (IAM), and ethical handling of business and customer information.

5. Feasibility and Practicality

Proposed solutions must be realistic considering cost, skill availability, and organizational readiness. Use of managed cloud services is encouraged instead of building everything from scratch.

6. Encouragement of Reflection

Students must include a short reflection covering:

How the solution relates to challenges in their own workplace
What cloud concepts they learned
What trade-offs were made and why

Hypothetical, abstract, or generic problems (e.g., "Online Shopping Website", "College Management System" without real context) will not be accepted.

Procedure:

- Apply theoretical knowledge to practical cloud implementations.
- Understand cloud architecture and service selection based on real-world use cases.
- Analyze system performance upgrades using measurable metrics.
- Develop collaboration skills through teamwork on problem-solving, design, and documentation.

Phase -1: Problem Statement Definition and Justification

For Phase 1:

Each group will be allocated 12 minutes for the presentation and 2 minutes for the Q&A session. If the chosen topic is not relevant to cloud computing, it will be rejected, and the group will be required to redo the presentation. Additionally, the group will lose marks for selecting an inappropriate topic.

Evaluation Rubrics for Phase - 1

CRITERIA	DESCRIPTION	MARKS ALLOCATION
Problem Title	The title clearly defines a Situation Learning (SL) based real-world problem within the scope of Cloud Computing domain..	1 Mark
Survey & Justification	-Conducts a brief survey/literature review of existing solutions. -Clearly explains the relevance and necessity of solving this problem in a cloud environment.	2 Marks
Objectives	- Defines clear and measurable objectives based on the stated Learning Objectives (mentioned above) and aligned with Situation Learning (SL) guidelines. - Ensures the problem is derived from a real-time workplace scenario and explains how cloud technologies can be used to achieve these objectives.	2 Marks
Milestones & Development Journey	- Breaks down the problem-solving process into distinct phases. -Allot duration for each phase	1 Marks
Requirements for Problem Solving	- Lists cloud services, tools, and technologies required to implement the solution	1 Mark
Roles & Responsibilities	- Clearly defines each member's role in solving the problem. - Ensures that all team members contribute meaningfully.	1 Mark

Guidelines:

If you are using **Service X** in a different cloud platform (such as Azure or GCP), you **must map it to the equivalent AWS Service Y** as the course demonstrates **the use of AWS**. This

mapping should be **clearly defined in Phase 1 itself** as part of the **problem statement and justification**.
For example:

- If you are using **Azure Virtual Machines**, map it to **AWS EC2**.
- If you are using **Azure Blob Storage**, map it to **AWS S3**.

This ensures that even if you implement the project on a non-AWS platform, the **concepts align with AWS services**. This mapping must be included in the **Phase 1 presentation**
This is a **group activity**, not an individual exercise. Each group must consist of **four members**, with clearly defined roles for each participant. During the demonstration, any member of the group may be asked questions, and they should be able to respond effectively. All team members must have a **comprehensive understanding of the entire project** to ensure a well-coordinated presentation. After the Phase 1 presentation, the Team Lead (one person from each group) should upload the Presentation document (Name of the file should be **CC_Project_ProblemDefinition<GroupNumber>)**

Phase -2: Implementation and Submission of Deliverables

For Phase 2:

Submission Requirement:

A. PPT submission explaining implementation details.

B. Recording of the working project demonstrating key functionalities.

C. Report submission, including service mapping for non-AWS platforms.

⌛ Late Submission Penalty:

If the submission is delayed beyond 2nd May, 20% marks will be deducted per day from 3rd May onwards.

Evaluation Rubrics for Phase - 2

CRITERIA	DESCRIPTION	MARKS ALLOCATION
PPT Submission	- Includes a structured explanation of the implementation process. - Clearly presents cloud service	1 Marks

	usage and configurations.	
Project Recording Submission	- Demonstrates a fully functional implementation of the project. - Showcases cloud integration, deployments, and key features.	1 Marks
Code/Scripting done	- Submission of Code/Scripting done in pdf format with proper comments.	2 Marks
Report Submission	Submits a well-structured report with the required format: Abstract, Survey, Problem Statement, Objectives, Outcomes, References (if used), and a Short Reflection section. The Short Reflection must include: i. How their solution relates to challenges in their own workplace. ii. What concept of cloud they learned about. iii. What trade-offs they made and why. This reflection deepens learning and connects theory to practice. If a non-AWS cloud is used, maps services to AWS equivalents.	7 Marks

Guidelines:

- **File Naming:** Zip the File and Name the submission files as **CC_Project_Implementation<GroupNumber>** for Phase 2 deliverables.
- **Cloud Service Mapping:** If a non-AWS cloud service is used (e.g., Azure, GCP), students must **map each service to its AWS equivalent** within the report.
- **Clarity and Accuracy:** The report should comprehensively explain all aspects of the implementation, ensuring alignment with the problem statement and objectives.
- **Proper Documentation:** The report must include **detailed descriptions, screenshots, and references** where applicable.
- **Timely Submission:** All deliverable s must be submitted by --/--/----.

- **Late Submission Penalty:** If the submission is delayed beyond **May 3, 20% of the marks will be deducted per day** from03/**May/2026 onwards**.

Note: Report format will be shared

Phase -3: Final Presentation

Each group will be allocated 12 minutes for the presentation and 3 minutes for the Q&A session. The demonstration must showcase the implementation of the proposed solution, aligning with the problem statement and objectives. If any deviation is found from the initially defined objectives, marks may be deducted accordingly.

For Phase 3:

CRITERIA	DESCRIPTION	MARKS ALLOCATION
Refined Problem Statement	- Updates and refines the problem statement based on feedback from Phase 1. - Ensures clarity and scope alignment.	1 Marks
Objective-Based Explanation with Role Assignments	- Each team member explains their specific contribution to achieving project objectives. All the objectives in phase-1 claimed were Implemented	3 Marks
Viva Voce	- Team members should be prepared to answer technical and conceptual questions related to their project. - Evaluates individual understanding and teamwork.	1 Marks

Guidelines:

- Equal Participation: Each team member must actively contribute to the presentation, ensuring a balanced distribution of speaking time.
- Comprehensive Knowledge: Every participant should be well-versed in all aspects of the project, not just their assigned role.
- Viva Questions: Questions may be directed at any member regarding any part of the project. Responses like "this was done by another member" will not be accepted.
- Team Coordination: Clear communication and teamwork during the presentation will be assessed. An additional 1 mark will be awarded for strong team understanding and seamless collaboration.

Grading Overview for Learners

General summary for the review criteria. Provide a brief explanation on how the assignment will be graded. Learners will use this as a guide when completing the tasks. Please be clear, detailed, and concise:

Phase 1: Problem Statement Definition and Justification (8 Marks)

Your submission in this phase will be evaluated based on the clarity and relevance of the **problem statement** within the scope of cloud computing. Key grading criteria include:

✅ **1 Mark - Problem Title:** Clearly defines a relevant problem related to cloud computing.
✅ 2 **Marks** -Survey & Justification: Provides a well-researched background and justification for a Situation Learning (SL) based real-world workplace or industry problem within the cloud computing domain.
✅ **2 Marks** - **Objectives:** Clearly states measurable objectives and explains how cloud computing services will address them.
✅ 1 **Marks** - **Milestones & Development Journey:** Defines distinct development phases with assigned timelines.
✅ **1 Mark** - **Requirements for Problem Solving:** Lists necessary cloud services and tools for implementation.
✅ **1 Mark** - **Roles & Responsibilities:** Clearly assigns tasks and ensures meaningful participation from all team members.

Note: Groups failing to choose a cloud-related problem will lose marks. Presentations should be well-structured and each member should contribute.

Phase 2: Implementation and Submission of Deliverables (11 Marks)

Your submission in this phase will be evaluated based on the successful **implementation of the proposed solution** and proper documentation of cloud services used.
✅ **1 Marks - PPT Submission:**

- Includes a **structured explanation** of the implementation process.
- Clearly presents **cloud service usage** and key configurations.

✅ 1 **Marks - Project Recording Submission:**

- Demonstrates a **fully functional implementation** of the project.
- Showcases **cloud integration, deployments, and key features**.

✅ **2 Marks - Code/Scripting done**

- Demonstrates a **fully functional Code for Validation**
- **Demonstrates the functionality with proper comments**

✅ **7 Marks - Report Submission:**

- Submits a **well-structured report** following the required format (**Abstract, Survey, Problem Statement, Objectives, Outcomes, References**).
- If a **non-AWS cloud** is used, clearly **maps services to AWS equivalents**.

Note:

- All deliverables must be submitted by 03-May-2026
- Late Submission Penalty: 20% of marks will be deducted per day from May 3 onwards.
- Each team member should contribute to the implementation and be prepared to explain their work.
- Zip and submit files as CC_Project_Implementation<GroupNumber>.

Phase 3: Final Presentation (6 Marks)

Your submission in this phase will be evaluated based on the **presentation quality, individual understanding, and ability to demonstrate project outcomes**.
✅ 1 Marks - Refined Problem Statement:

- Updates and refines the **problem statement** based on feedback from **Phase 1**.
- Ensures **clarity and alignment** with the project scope.

✅ 3 Marks - Objective-Based Explanation with Role Assignments:

- Each team member explains their **specific contribution** to achieving the project objectives.
- Demonstrates that **all objectives defined in Phase 1 were successfully implemented**.

Provides supporting screenshots and explanations for each implemented feature.
✅ 1 Marks - Viva Voce:

- Team members must be prepared to answer **technical and conceptual questions** related to their project.
- Evaluates **individual understanding, problem-solving ability, and teamwork**.

Note:

- **Each group will be allocated 12 minutes for the presentation and 3 minutes for Q&A.**
- **Equal Participation:** Every team member must actively contribute to the presentation.
- **Viva Questions:** Questions may be directed at any member regarding any part of the project. Responses like *"this was done by another member"* will not be accepted.
- **Team Coordination: An additional 1 mark** will be awarded for **strong team understanding and seamless collaboration** during the presentation.

Grader Instructions

Optional: General summary for the grading criteria (only visible to graders).
The **Cloud Computing project** will be graded based on structured **evaluation rubrics** for each phase. The submission will be primarily assessed based on: • **Clarity and relevance** of the problem statement within the cloud computing domain. • **Proper justification, documentation, and implementation** of cloud services. • **Balanced participation among team members** in the presentation. • **Accuracy in demonstrating project outputs** and providing complete deliverables. • **Strict adherence to the provided report format** and evaluation guidelines.

Important:

✔ **Every student in a group must have a complete understanding of the entire project.**

✔ **Marks will be deducted for incomplete, inconsistent, or poorly documented work.**

Grading Criteria & Rubrics

Rubric 1 – Phase 1: Problem Statement Definition and Justification (8 Marks)

✔ Did the group define a relevant cloud computing problem statement with a justified survey and objectives?
✔ Are the milestones well-defined, and are the required cloud services listed?
✔ Are team roles and responsibilities clearly mentioned?
✔ If using a non-AWS cloud, did they provide service mapping to AWS?

Max Points: 8

Grading Type: Presentation + PPT Submission

Rubric 2 – Phase 2: Implementation and Submission of Deliverables (11 Marks)

✔ Did the group submit a well-structured PPT explaining the implementation process?
✔ Did the group submit Code/ Scripts with proper comments
✔ Did the project recording showcase a fully functional implementation with cloud service integration?
✔ Did the report follow the required format and include AWS service mapping if using a non-AWS cloud?
✔ Was the submission made before the deadline (03-05-2026)?
✔ If late, was the appropriate penalty applied (20% mark deduction per day from May 3 onwards)?

Max Points: 11

Grading Type: PPT + Project Recording + Report Submission

Rubric 3 – Phase 3: Final Presentation & Q&A (6 Marks)

✔ Did the group refine the problem statement based on Phase 1 feedback?
✔ Did each team member explain their specific contribution clearly?

✔ Did the group successfully implement all objectives from Phase 1?
✔ Did the project output demonstration provide evidence of execution through screenshots and results?
✔ Did all team members actively participate in answering viva questions?
✔ Did the team demonstrate strong coordination and seamless collaboration?

Max Points: 6

Grading Type: Presentation + Q&A Session